웹 배포 가이드

배포 방법

1. 환경 구성
    a. Ubuntu 22.04 LTS 환경에 Jenkins 설치
    b. Jenkins Web GUI 접속(기본 포트 8080)하여 초기 설정 진행
        - 계정 정보

          id : admin

          password :K12A402T

          이름 : ssafy

          email : k12a402@ssafy.com

          ```
          /var/lib/jenkins
          ```

    c. 필요한 credential 값들 Jenkins credentials에 입력 (하단의 환경 변수 설명에서 확인 가능)
    d. nginx 설치 & certbot SSL 적용
        - 설정 파일 구성

          /etc/nginx/sites-available/
          ├── default.conf # catch-all 또는 리디렉션 용도
          ├── leafree.conf # leafree.kr, www.leafree.kr → leafree.kr
          ├── jenkins.conf # jenkins.leafree.kr → Jenkins 서버

        ```
        sudo letsencrypt certonly --standalone -d leafree.kr -d www.leafre
        e.kr
        ```

        ```
        server {
            listen 80;
            server_name leafree.kr www.leafree.kr test.leafree.kr;
            return 301 https://$host$request_uri;
        }
        ```

```
server {
        listen 443 ssl http2;
        server_name leafree.kr;

        ssl_certificate /etc/letsencrypt/live/leafree.kr/fullchain.pem;
        ssl_certificate_key /etc/letsencrypt/live/leafree.kr/privkey.pem;

        root /var/www/leafree.kr;
        index index.html;

        location / {
                try_files $uri $uri/ /index.html;
        }

        location /downloads/ {
                alias /var/www/leafree.kr/downloads/;
        }

        location /api/ {
                proxy_pass http://127.0.0.1:9000/api/;
                proxy_set_header Host $host;
                proxy_set_header X-Real-IP $remote_addr;
                proxy_set_header X-Forwarded-For $proxy_add_x_forwar
ded_for;
                proxy_set_header X-Forwarded-Proto $scheme;

        }

        location /llm-api/ {
                proxy_pass http://127.0.0.1:8000/;
                proxy_set_header Host $host;
                proxy_set_header X-Real_IP $remote_addr;
                proxy_set_header X-Forwarded-For $proxy_add_x_forwar
ded_for;
                proxy_set_header X-Forwarded-Proto $scheme;
        }
}
```

```
server {
	listen 443 ssl http2;
	server_name www.leafree.kr;

	ssl_certificate /etc/letsencrypt/live/leafree.kr/fullchain.pem;
	ssl_certificate_key /etc/letsencrypt/live/leafree.kr/privkey.pem;

	return 301 https://leafree.kr$request_uri;
}

server {
	listen 443 ssl http2;
	server_name test.leafree.kr;

	ssl_certificate /etc/letsencrypt/live/jenkins.leafree.kr/fullchain.
pem;
	ssl_certificate_key /etc/letsencrypt/live/jenkins.leafree.kr/privk
ey.pem;

	root /var/www/test.leafree.kr;
	index index.html;

	location / {
		try_files $uri $uri/ =404;
	}
}
```

```
sudo ln -s /etc/nginx/sites-available/leafree.conf /etc/nginx/sites-e
nabled/
```

e. Redis 설치 & 설정

```
sudo vi /etc/redis/redis.conf
```

```
sudo service redis-server restart
```

```
# 원래 주석처리 되어있음. 최대 메모리 2gb로 제한
maxmemory 2gb
```

```
# 원래 주석처리 되어있음. 자주 사용되지 않은 키부터 자동 제거
maxmemory-policy allkeys-lru
```

```
# 완전 휘발성. 캐시 용도
save ""
# save 900 1
# save 300 10
# save 60 10000
```

f. MySQL 설치 & 설정

```
docker exec -it local-mysql mysql -u root -p
```

```
ALTER USER 'root'@'%' IDENTIFIED WITH mysql_native_password BY 'yourpassword';
```

```
ALTER USER 'k12a402'@'%' IDENTIFIED WITH mysql_native_password BY 'K!@A$)@';
```

g. Mosquitto 설치 & 설정

```
sudo apt update
sudo apt install mosquitto mosquitto-clients -y
```

```
sudo systemctl enable mosquitto
sudo systemctl start mosquitto
```

```
# Place your local configuration in /etc/mosquitto/conf.d/
#
# A full description of the configuration file is at
```

```
# /usr/share/doc/mosquitto/examples/mosquitto.conf.example

pid_file /run/mosquitto/mosquitto.pid

persistence true
persistence_location /var/lib/mosquitto/

log_dest file /var/log/mosquitto/mosquitto.log

include_dir /etc/mosquitto/conf.d

listener 1883
protocol mqtt
# bind_address 0.0.0.0

allow_anonymous false
password_file /etc/mosquitto/passwd
acl_file /etc/mosquitto/acl
```

```
user testuser
topic readwrite #

user android_user
topic read /leafree/+/sensor/#
```

h. ufw 설정

- 포트 정보

  외부

  - 80 : http 연결 → 443
  - 1883 : mqtt 연결

  내부

  - 8080 : jenkins
  - 9000 : 스프링부트 서버
  - 8000 : FastAPI 서버

    - 1883 : Mosquitto
    - 3306 : MySQL
    - 6379 : Redis

구성 요소 별 환경설정

- **IDE**
    - Intellij IDEA 2024.3.1.1
    - Android Studio 2024.3.1 Patch 2
- **웹 서버**

항목	설명
Nginx	리버스 프록시 및 정적 파일 서빙용

- 백엔드 서버

항목	값 또는 설명	버전
언어	Java	17
프레임워크	Spring Boot	3.4.5
포트	`9000`	
DB 연동	MySQL DB	8.0.42
인증 방식	JWT	0.11.5
MQTT 브로커	Mosquitto	2.0.11
캐시	Redis	6.0.16

    - 의존성 목록
        - org.springframework.boot:spring-boot-starter-web
        - org.springframework.boot:spring-boot-starter-webflux
        - org.springframework.integration:spring-integration-webflux
        - org.springframework:spring-web
        - org.springframework.boot:spring-boot-devtools
        - org.springframework.boot:spring-boot-starter-data-jpa
        - com.mysql:mysql-connector-j (runtimeOnly)

        - org.springframework.boot:spring-boot-starter-security
        - org.springframework.boot:spring-boot-starter-oauth2-client
        - io.jsonwebtoken:jjwt-api:0.11.5
        - io.jsonwebtoken:jjwt-impl:0.11.5
        - io.jsonwebtoken:jjwt-jackson:0.11.5
        - org.springframework.boot:spring-boot-starter-data-redis
        - org.springframework.integration:spring-integration-redis
        - org.springframework.boot:spring-boot-starter-integration
        - org.springframework.integration:spring-integration-mqtt
        - org.eclipse.paho:org.eclipse.paho.client.mqttv3
        - org.springdoc:springdoc-openapi-starter-webmvc-ui:2.8.8
        - org.springframework.boot:spring-boot-starter-test
        - org.springframework.security:spring-security-test
        - io.projectreactor:reactor-test
        - org.junit.platform:junit-platform-launcher (testRuntimeOnly)
        - org.springframework.boot:spring-boot-starter-validation
        - org.springframework.boot:spring-boot-starter-actuator
        - org.projectlombok:lombok (compileOnly + annotationProcessor)
        - com.google.code.gson:gson:2.8.9
        - com.fasterxml.uuid:java-uuid-generator:4.0.1
        - com.google.firebase:firebase-admin:9.4.3
- 모바일

항목	설명
언어	Kotlin
UI 프레임워크	Jetpack Compose + Material3
상태관리	`remember` + `ViewModel` 사용
알림 기능	Firebase Cloud Messaging (FCM)

    - 의존성 목록

- androidx.compose:compose-bom:2024.05.00
- androidx.compose.material3:material3
- androidx.compose.foundation:foundation:1.8.1
- androidx.compose.ui:ui
- androidx.compose.ui:ui-tooling-preview
- androidx.compose.ui:ui-tooling
- androidx.compose.ui:ui-test-junit4
- androidx.compose.ui:ui-test-manifest
- androidx.compose.material:material-icons-core
- androidx.compose.material:material-icons-extended
- androidx.compose.material3.adaptive:adaptive:1.0.0-beta03
- androidx.activity:activity-compose:1.10.1
- androidx.lifecycle:lifecycle-viewmodel-compose:2.8.5
- androidx.navigation:navigation-compose:2.9.0
- io.coil-kt.coil3:coil-compose:3.2.0
- io.coil-kt.coil3:coil-network-okhttp:3.2.0
- org.jetbrains.kotlinx:kotlinx-coroutines-android:1.8.0
- com.squareup.retrofit2:retrofit:2.9.0
- com.squareup.retrofit2:converter-gson:2.9.0
- com.squareup.okhttp3:okhttp:4.12.0
- com.squareup.okhttp3:logging-interceptor:4.12.0
- com.kizitonwose.calendar:compose:2.7.0
- org.eclipse.paho:org.eclipse.paho.client.mqttv3:1.2.5
- io.github.mayzs:paho.mqtt.android:1.2.1
- com.google.firebase:firebase-bom:33.13.0
- com.google.firebase:firebase-analytics
- com.google.firebase:firebase-messaging

    - org.junit.jupiter:junit-jupiter:5.12.2
    - org.junit.platform:junit-platform-launcher
- 기타

항목	설명
MQTT 통신 방식	IoT → MQTT → Spring Boot & Android
Firebase	FCM 알림 발송, Admin SDK 인증 JSON 사용
CI/CD	Jenkins + GitLab + Docker

CICD 파이프 라인

- gitlab 의 develop-web 에 push가 발생할 경우 Jenkins가 Webhook으로 감지
- develop-web의 Web/backend 폴더의 Jenkinsfile 을 통해 빌드, 배포 진행
    - jenkins credentials

ID	설명 또는 저장된 값
`gitlab-token`	GitLab Personal Token
`gitlab-api-token`	GitLab API token
`JWT_SECRET`	JWT 서명 키
`DB_PASSWORD`	MySQL DB 비밀번호
`MQTT_CLIENT_ID`	MQTT 클라이언트 ID
`MQTT_USERNAME`	MQTT 사용자 이름
`MQTT_PASSWORD`	MQTT 비밀번호
`FIREBASE_CREDENTIAL_JSON`	Firebase Admin SDK 인증 파일 이름 (`firebase-adminsdk.json`)

    - JenkinsFile

```
pipeline {
    agent any

    environment {
        IMAGE_NAME = "leafree-backend"
        CONTAINER_NAME = "leafree-backend"

        // 실제 환경변수들
```

```
        JWT_SECRET = credentials('JWT_SECRET')

        DB_PASSWORD = credentials('DB_PASSWORD')

        MQTT_CLIENT_ID = credentials('MQTT_CLIENT_ID')
        MQTT_USERNAME = credentials('MQTT_USERNAME')
        MQTT_PASSWORD = credentials('MQTT_PASSWORD')

        FIREBASE_CREDENTIAL_JSON = credentials('FIREBASE_CRED
ENTIAL_JSON')
    }

    stages {
        stage('Checkout') {
            steps {
                checkout scm
            }
        }

        stage('Build') {
            steps {
                dir('Web/backend') {
                    sh './gradlew clean build -x test'
                }
            }
        }

        stage('Docker Build') {
            steps {
                dir('Web/backend') {
                    sh '''
                        docker build -t $IMAGE_NAME .
                    '''
                }
            }
        }

        stage('Deploy (local)') {
```

```
        steps {
            sh '''
                docker stop $CONTAINER_NAME || true
                docker rm $CONTAINER_NAME || true
                docker run -d --network="host" --name $CONTAINER_
NAME \
                  -e JWT_SECRET=$JWT_SECRET \
                  -e DB_PASSWORD=$DB_PASSWORD \
                  -e MQTT_CLIENT_ID=$MQTT_CLIENT_ID \
                  -e MQTT_USERNAME=$MQTT_USERNAME \
                  -e MQTT_PASSWORD=$MQTT_PASSWORD \
                  -e GOOGLE_APPLICATION_CREDENTIALS=/app/fireba
se-key.json \
                  -v $FIREBASE_CREDENTIAL_JSON:/app/firebase-key.j
son:ro \
                  -p 9000:9000 \
                  $IMAGE_NAME
            '''
        }
    }
  }

  post {
    success {
      echo "✅ 배포 완료"
    }
    failure {
      echo "❌ 배포 실패"
    }
  }
}
```

DB 접속 정보

- 접속 정보 (ssh 키 필요)

항목	값
DB_HOST	i12a106.p.ssafy.io

DB_PORT	8999
DB_NAME	masterdb
DB_USERNAME	ssafy
DB_PASSWORD	(Jenkins Credential 사용)

- ddl

```
CREATE TABLE users (
    user_id BIGINT NOT NULL AUTO_INCREMENT,
    user_email VARCHAR(256) NOT NULL UNIQUE,
    user_password VARCHAR(256) NOT NULL,
    user_nickname VARCHAR(256) NOT NULL,
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (user_id)
);

CREATE TABLE devices (
    device_id BIGINT NOT NULL AUTO_INCREMENT,
    device_uuid CHAR(36) NOT NULL UNIQUE,
    created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (device_id)
);

CREATE TABLE user_devices (
    user_id BIGINT,
    device_id BIGINT,
    created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (user_id, device_id)
);

CREATE TABLE crops (
    crop_id BIGINT NOT NULL AUTO_INCREMENT,
    device_id BIGINT NOT NULL,
    crop_type VARCHAR(50) NOT NULL,
    crop_status VARCHAR(20), -- GROWING,WARNING,HARVESTABLE
    crop_image LONGBLOB,
    created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
    harvested_at DATE,
```

```
    PRIMARY KEY (crop_id)
);

CREATE TABLE sensor_logs (
    log_id BIGINT NOT NULL AUTO_INCREMENT,
    device_id BIGINT NOT NULL,
    temperature FLOAT,
    humidity INT,
    light INT,
    co2_level INT,
    water_level INT,
    soil_moisture INT,
    logged_at DATETIME DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (log_id)
);

CREATE TABLE device_settings (
    device_id BIGINT NOT NULL PRIMARY KEY,  -- PK로 변경
    optimal_temperature FLOAT,            -- 최적 온도 (°C)
    optimal_humidity INT,                 -- 최적 습도 (%)
    optimal_light INT,                    -- 최적 조도 (lux)
    optimal_co2_ppm INT,                  -- 최적 CO2 농도 (ppm)
    optinal_soil_humidity INT,            -- 최적 토양 수분 (%)
    light_duration INT,                   -- 조도 유지 시간 (분 단위, 08:00부터 시
작)
    watering_interval INT,                -- 물 주기 (몇 분마다)
    updated_at DATETIME DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP,
    FOREIGN KEY (device_id) REFERENCES devices(device_id) ON DELE
TE CASCADE
);

CREATE TABLE events (
    event_id BIGINT AUTO_INCREMENT PRIMARY KEY,
    device_id BIGINT NOT NULL,
    event_type VARCHAR(50) NOT NULL,       -- ACTUATING, WARNING,
ERROR, NOTIFY
    event_content_code VARCHAR(50) NOT NULL, -- "TEMP_CHANGE
```

```
D", "DEVICE_ERROR"
    event_data JSON,                    -- {"from":22.5, "to":25.0}
    occurred_at DATETIME NOT NULL,
    FOREIGN KEY (device_id) REFERENCES devices(device_id)
);

CREATE TABLE fcm_tokens (
    token_id BIGINT NOT NULL AUTO_INCREMENT PRIMARY KEY,
    user_id BIGINT NOT NULL,
    fcm_token VARCHAR(256) NOT NULL UNIQUE,
    updated_at DATETIME DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP,
    created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE CA
SCADE
);

-- user_devices.device_id → devices.device_id
ALTER TABLE user_devices
ADD CONSTRAINT fk_user_devices_device
FOREIGN KEY (device_id)
REFERENCES devices(device_id)
ON DELETE CASCADE;

-- user_devices.user_id → users.user_id
ALTER TABLE user_devices
ADD CONSTRAINT fk_user_devices_user
FOREIGN KEY (user_id)
REFERENCES users(user_id)
ON DELETE CASCADE;

-- crops.device_id → devices.device_id
ALTER TABLE crops
ADD CONSTRAINT fk_crops_device
FOREIGN KEY (device_id)
REFERENCES devices(device_id)
ON DELETE CASCADE;
```

```
-- sensor_logs.device_id → devices.device_id
ALTER TABLE sensor_logs
ADD CONSTRAINT fk_sensor_logs_device
FOREIGN KEY (device_id)
REFERENCES devices(device_id)
ON DELETE CASCADE;

CREATE INDEX idx_event_device ON events(device_id);
CREATE INDEX idx_event_type ON events(event_type);
CREATE INDEX idx_event_time ON events(occurred_at);
CREATE INDEX idx_event_device_type_time ON events(device_id, event
_type, occurred_at DESC);
```